# Bounded-memory streaming redaction of personal data and credentials with a whole-document equivalence guarantee

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**Abstract**

Logs and data exports carry personal data and credentials and often exceed memory. Chunked processing adds a silent failure: a value straddling a chunk boundary reaches the detector in two halves and is written out in the clear. SafeStream-Redactor treats chunking as a correctness problem. Each detector declares its longest span and the context it reads. We show that an overlap of the longest span can still leak, and prove that if the overlap covers every detector's span plus its right context, the streamed output equals a whole-document pass, parallel output equals sequential output or falls back to it, and the working buffer is independent of input size; the engine enforces this overlap. Naive 1 KiB chunking leaked 237 RedactionBench values a whole-document pass redacts and missed 388 in real logs; overlapping windows still missed some, SafeStream none. Streaming added 8.4–8.54 MiB from 8 MiB to 2 GiB, against 904.6 MiB for whole-document redaction of 32 MiB. On 14,626 real credentials labelled by people (CredData), the pattern tiers fully redacted 0.55 against 0.40 for gitleaks, robust to resampling by repository, at a higher false-alarm rate. An opt-in tier caught all 618 user names in held-out real Linux logs. On 1,014 held-out real court judgments, a free-text configuration frozen before scoring reached character F1 0.67 against 0.54 for Presidio and fully redacted 89% of direct identifiers against 13%. Yet no configuration fully protected more than 17.9% of RedactionBench documents.

**Keywords:** Data redaction, Secret detection, Data loss prevention, Streaming algorithms, Bounded memory, Personally identifiable information

# 1 Introduction

Text artefacts that leave an organisation, such as an application log attached to a support ticket, a database export handed to an analyst, or a packet capture shared with a vendor, routinely carry two classes of sensitive value: personal data (e-mail addresses, telephone and card numbers, national identifiers, names) and credentials (application programming interface keys, access tokens, private keys, passwords). Data-loss-prevention systems inspect such content before it crosses a boundary [1], and measurement studies show that credentials leak into shared artefacts at scale [2, 3].

Two families of tools address the problem. Personal-data engines such as Microsoft Presidio [4] combine pattern recognisers with a statistical named-entity model but ship no credential recognisers. Secret scanners such as gitleaks [5] and TruffleHog [6] target credentials in source repositories, and their accuracy varies widely across secret types [7]. The artefacts that most need redaction are often too large to analyse in one call, so they are processed in pieces. A chunked redactor can then fail silently: a value that straddles a chunk boundary is presented to the detector in two halves, neither of which matches, and both halves are written out in the clear. The same problem is long established in network security, where signatures split across packets motivated stream reassembly in intrusion detection [8] and streaming modes in regular-expression engines [9, 10]. In redaction, however, detectors read context around a match ("Patient ID:" before a number, a key name before a token), confidences are adjusted by neighbouring text, and overlapping detections are merged before the text is rewritten. Overlapping windows are the obvious remedy, but how much overlap is enough depends on all of these, and the obvious answer, an overlap as long as the longest match, is wrong: a match whose right context is cut off at the window edge can be rejected there, so the boundary does not retreat out of it and its first part is emitted in the clear. None of the redaction or secret-scanning systems we examined states when its chunked output equals its whole-document output.

This paper answers that question for a concrete system, SafeStream-Redactor (hereafter SafeStream). Its contributions are the following.

1. **A tight, per-detector overlap condition for context-dependent redaction.** Every detector $d$ declares an upper bound $L_d$ on the span it can emit and bounds $C_{\mathrm{left},d}$ and $C_{\mathrm{right},d}$ on the context it reads (Assumptions A1 and A2). We prove that an overlap $w \geq \max_d \max(L_d + C_{\mathrm{right},d}, C_{\mathrm{left},d})$ makes the streamed output equal to the whole-document output for every chunking, and show that the term $L_d + C_{\mathrm{right},d}$ cannot be dropped: an overlap of $\max_d L_d$, the natural choice, can leak (Section 5). We also measure the failure the condition prevents: on the test corpora, naive chunking leaked up to 237 gold values that a whole-document pass redacts, and overlapping windows still leaked some, while SafeStream's output was identical to the whole-document output under every chunking tried (Section 7.1).

The underlying locality argument is standard; what is new is the condition for pipelines whose detectors read context on both sides, adjust confidences and resolve overlaps jointly, which pattern-matching engines that carry automaton state across blocks do not address.

2. **A guarantee enforced by the code, not chosen by hand.** The engine computes the condition from the detectors' declarations and refuses to run with a smaller overlap; regression and differential tests check that the declarations hold. For the default pipeline the requirement is exactly the default overlap of 8192 characters.
3. **Parallel redaction that cannot lose or duplicate text.** A seam protocol between worker processes makes the parallel output equal to the sequential output under the same condition, and falls back to sequential processing on any disagreement, so no character is lost or emitted twice whatever the detectors do (Appendix A).
4. **Bounded memory, measured.** The working buffer is bounded independently of input size, and peak memory stays flat across a 256-fold range of input sizes, where whole-document redaction needs about 28 times the input size (Section 7.2).
5. **A detection evaluation that separates robust from fragile margins.** Equivalence says nothing about how good the detectors are. We evaluate them on held-out test splits against Presidio and two secret scanners, with paired bootstrap intervals at the document, cluster and two-stage levels, a scanner sensitivity run with false-positive filters disabled, an ablation and a contamination analysis (Sections 6 and 7). Two margins survive every check: a higher fully redacted rate for credentials than gitleaks, confirmed on real credentials from 337 repositories at the price of more false alarms, and, on a synthetic corpus, higher character F1 than Presidio with the same spaCy model. Others do not. On real court judgments annotated by the Text Anonymization Benchmark the second margin reverses for the evaluated configuration; a free-text configuration designed on that benchmark's development split and confirmed on 1,014 held-out judgments detects substantially more personal data than Presidio (Section 7.9).

Two scope limits are stated at the outset. First, the guarantee concerns equivalence rather than accuracy: streaming does not improve detection, it makes the streamed result the same as the whole-document result, so every detection error of the whole-document pipeline is also an error of the streamed one. Second, the evaluation has known weaknesses that we quantify rather than omit: one corpus, RedactionBench, was inspected during rule development (Section 7.5); two corpora are synthetic, and on real court judgments the evaluated named-entity configuration detected less personal data than Presidio until a free-text configuration was added (Section 7.8); and fewer than one RedactionBench document in five was fully protected by any configuration we evaluated (Section 7.5). Section 8 states when the approach is useful despite these limits and what evidence a deployment would need.

## 2 Background and related work

***Personal-data detection and de-identification***

Rule-based de-identification of clinical text dates back to lookup tables and regular expressions applied to nursing notes [11]; Philter refined the approach on a large annotated clinical corpus [12]. The 2014 i2b2/UTHealth shared task established annotated corpora and evaluation practice [13], and recurrent neural networks reached strong results on them [14]. Presidio [4] combines pattern recognisers with spaCy or transformer models and ships an evaluation toolkit with a synthetic dataset [15]. Lison et al. argue that sequence labelling alone does not amount to anonymisation and call for explicit disclosure-risk measures [16]; the Text Anonymization Benchmark operationalises this for court judgments [17]. Named-entity recognition underpins the free-text part of these systems, from the CoNLL-2003 shared task [18] to transformer encoders [19, 20] and generalist span models [21]. Pseudonymisation with a keyed hash such as HMAC [22] preserves referential integrity but reveals equality between values, so it is not anonymisation in the sense of $k$-anonymity [23]. Redaction as studied here is narrower than anonymisation: it removes spans that detectors recognise and makes no claim about re-identification from what remains.

***Credential and secret detection***

Meli et al. measured secret leakage across public GitHub repositories [2]. Approaches range from vendor-specific regular expressions [24] to machine-learned filters that reduce false positives [25] and neural detectors for free-form passwords [26]. SecretBench provides a labelled corpus of candidate secrets [27], and a comparison of nine tools attributed false positives to generic patterns and entropy heuristics and false negatives to missing rules [7]. Krause et al. studied how developers handle leaked secrets [3]. Shannon entropy [28] is the usual signal for undocumented high-entropy secrets. The Nosey Parker rule set [29] supplies the labelled examples we evaluate on.

***Detection across segment boundaries***

That a pattern split across units of input can evade a detector is well known in network security: Ptacek and Newsham showed that intrusion detectors which do not reconstruct streams as the end host does can be evaded [8], and Hyperscan [9] provides a streaming mode whose matches do not depend on how the data is split into blocks [10]. These systems carry automaton state across boundaries, which makes pattern matching exact but says nothing about what happens after a match: validators, context-dependent confidences and joint overlap resolution all read text the automaton has already consumed. A redaction pipeline differs in three ways that affect correctness: detectors consult bounded context on both sides of a match, confidences depend on nearby text, and overlapping detections are resolved jointly before the output is rewritten. Overlapping windows are a common way to process long inputs, and the stream model [30] formalises computation in bounded memory; linear-time regular-expression matching goes back to Thompson [31]. Among the redaction, de-identification and secret-scanning systems cited above, we found none that states when chunked output equals whole-document output or how the overlap must be sized. We

do not claim that the underlying locality argument is new; the contribution is its instantiation and enforcement for context-dependent redaction.

# 3 Problem statement and threat model

**Definition 1** (Detections and redaction) A text $T$ is a finite character string, and $T[a{:}b]$ is its half-open substring. A *detection* is a tuple $(i, j, \tau, c)$ with $0 \le i < j \le |T|$: a half-open span, an entity type and a confidence in $[0, 1]$. Each detection is produced by one detector $d$ of a finite set $\mathcal{D}$. A *pipeline* $\Pi$ maps $T$ to a set of pairwise-disjoint detections. *Redaction* rewrites $T$ by replacing each span in $\Pi(T)$ according to a policy (replace, mask, hash, keyed pseudonym or encryption). The replacement of a span depends only on its type and its text, so redaction is a deterministic function of $T$ and $\Pi(T)$ that acts on each span separately.

**Definition 2** (Leak) Given a set $G$ of gold sensitive spans for $T$, a *leak* is any character of a span in $G$ that is not covered by a span in $\Pi(T)$. A gold span with at least one uncovered character is *leaked*, because a partly visible card number or key can often be reconstructed or narrowed.

### *Adversary and setting*

The adversary is a later reader of the redacted output, for example the recipient of a shared log. The input may be larger than memory and arrive as an arbitrary sequence of chunks. The system runs locally and makes no network calls, so the unredacted text does not leave the host.

### *What is in scope*

(i) That chunked and parallel processing introduce no leak beyond those of the whole-document pipeline; (ii) bounded working memory; (iii) the detection quality of the whole-document pipeline, measured empirically.

### *What is out of scope*

Detection completeness: the pipeline misses values its detectors do not recognise, and the guarantee does not change that. Adversarial evasion: an author who encodes, splits or obfuscates a secret on purpose is not modelled. Inference from residual text: quasi-identifiers and context that remain after redaction may still allow re-identification [16, 23]. Side channels such as output length. Bounded memory is a resource property, not a security property.

# 4 Design

Figure 1 shows the architecture. The reference pipeline $\Pi$ runs every configured detector on a text, adjusts confidences with a bounded contextual scorer, drops detections below a threshold (0.5 by default) and resolves overlaps into a disjoint set.

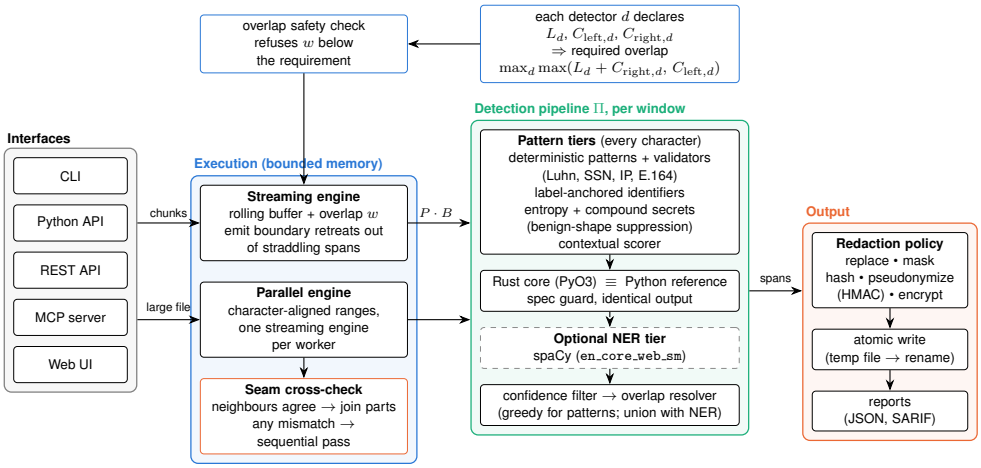



**Fig. 1** System architecture. Interfaces feed text to the streaming engine or, for a large file, to the parallel engine, whose workers each run a streaming engine and whose seams are cross-checked. Each window passes through the detection pipeline $\Pi$: the pattern tiers run on every character, on the Rust core or the identical Python reference. The dashed named-entity tier (spaCy) is optional. The resolver produces disjoint spans, which the redaction policy rewrites and an atomic write commits. The overlap $w$ is checked against the span and context bounds the active detectors declare.

## 4.1 Detection tiers

### *Deterministic patterns with validators*

Credential formats with a fixed shape (cloud access keys, provider tokens, JSON Web Tokens, PEM private-key blocks) and structured personal data (e-mail, telephone, card and social-security numbers, IPv4 and IPv6 addresses) are matched by regular expressions. A match becomes a detection only if it passes a validator: the Luhn checksum for card numbers [32], area, group and serial rules for social-security numbers, octet ranges for IPv4, address parsing for IPv6, and an E.164 shape check for telephone numbers. A generic key-assignment rule matches values that follow keys such as `api_key`, `token` or `password`.

### *Label-anchored identifiers*

Unchecked numeric identifiers such as medical record, account or licence numbers are ambiguous in isolation, so this tier promotes a value to a detection only when a cue word that names it (for example `Patient ID:`) appears within 96 characters to its left.

### *Entropy tier with benign-shape suppression*

Unstructured candidate runs of 20 to 8192 characters over a key alphabet are scored by length, character-class mix and Shannon entropy [28], with a lower threshold for hexadecimal runs; a compound variant covers runs with embedded symbols. Entropy alone cannot separate a secret from other random-looking strings. During this study

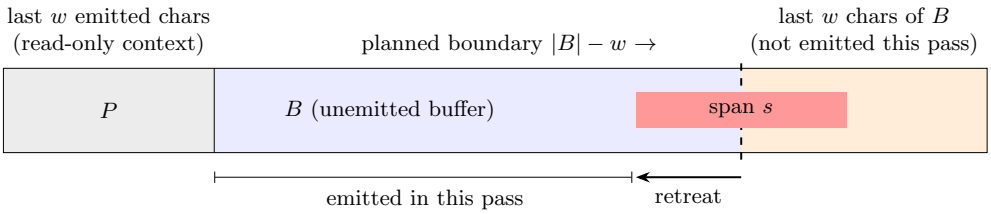



**Fig. 2** One pass of Algorithm 1. The detector reads $P \cdot B$; only text before the planned boundary may be emitted, and the boundary retreats out of any span that crosses it, so $s$ is carried to the next pass and detected whole.

every generated UUID, commit hash, SHA-256 digest and base64 image payload in the secrets development split was flagged (Section 7.4). The tier therefore suppresses three shapes: a canonical UUID, a 40- or 64-character hexadecimal value preceded by a digest cue (`commit`, `sha256`, `hash` and similar), and the payload of a `data:` URI. Suppression is vetoed when a secret-like key (`key`, `token`, `secret`, `password` and similar) names the value, so `SAUCE_ACCESS_KEY=`*uuid* remains a secret. The check reads at most 32 characters of left context.

#### *Contextual scorer*

A 48-character left window raises or lowers confidences when trigger words are present and emits a small set of trigger-gated detections of its own.

#### *Optional named-entity tier*

For names, organisations and locations, an optional tier runs a spaCy pipeline (`en_core_web_sm` in our experiments).

## 4.2 Overlap resolution

Pattern detections that overlap are resolved best-first by confidence, then by length, so a card number beats a telephone number embedded in it. Named-entity detections are handled differently: any cluster of overlapping spans that contains a named-entity span becomes one span from the cluster's leftmost start to its rightmost end, labelled by its best pattern member if one exists. Without this union rule, a PERSON tag on `john.doe` could replace the e-mail detection for `john.doe@example.com` and leave the domain exposed.

## 4.3 Streaming engine

Algorithm 1, a simplified form of `engine.stream_windows` illustrated in Figure 2, keeps a buffer of unemitted text and the last $w$ characters already emitted. Each pass detects over their concatenation, so detectors see real left context rather than an artificial edge, and emits only the part of the buffer that lies at least $w$ characters before its end. If a detection crosses the planned emit boundary, the boundary retreats to the detection's start and the detection is carried into the next pass. Small chunks

**Algorithm 1** Streaming detection with overlap $w$

**Require:** chunks $c_1, c_2, \ldots$; pipeline $\Pi$; overlap $w$; minimum emit size $m \geq w$

$P \leftarrow \varepsilon$ ▷ emitted text kept as read-only left context
$B \leftarrow \varepsilon$ ▷ buffer of unemitted text
**for** each chunk $c$ **do**
  $B \leftarrow B \cdot c$
  **if** $|B| < w + m$ **then**
    **continue**
  **end if**
  $S \leftarrow \{(i - |P|,\ j - |P|,\ \tau,\ \kappa) : (i, j, \tau, \kappa) \in \Pi(P \cdot B),\ i \geq |P|\}$
  $e \leftarrow |B| - w$ ▷ planned emit boundary
  **while** some $(i, j, \tau, \kappa) \in S$ has $i < e < j$ **do**
    $e \leftarrow i$ ▷ retreat out of the straddler
  **end while**
  **if** $e \leq 0$ **then**
    **continue** ▷ a span pins the buffer; read more
  **end if**
  **emit** $B[0{:}e]$ with $\{(i, j, \tau, \kappa) \in S : j \leq e\}$
  $P \leftarrow$ last $w$ characters of $P \cdot B[0{:}e]$
  $B \leftarrow B[e{:}]$
**end for**
**emit** $B$ with $\{(i - |P|,\ j - |P|,\ \tau,\ \kappa) : (i, j, \tau, \kappa) \in \Pi(P \cdot B),\ i \geq |P|\}$

are coalesced until at least $m$ characters can be emitted, which bounds the cost of rescanning the overlap per emitted character regardless of how the input is chunked. The defaults are $w = 8192$ characters and $m = 65{,}536$ characters.

## 4.4 Parallel engine

For large files the parallel engine splits the input into character-aligned byte ranges of at least $4w$ characters, one per worker process. Each worker reads $w$ characters of scanned padding and a further $w$ characters of context-only prefix around its interior boundaries (seams). A span that crosses a seam belongs to the left worker, and the right worker begins emitting at that span's end. Neighbouring workers compute the seam position independently, and the coordinator compares their reports; on any disagreement it discards the parallel result and redacts the file sequentially.

## 4.5 Implementation

The pure-Python implementation is the reference. An optional Rust core implements the pattern tiers. At start-up it checks that its compiled pattern sources equal the Python ones, and it hands a window back to Python in the rare cases where Unicode case folding or floating-point summation could make the two disagree.

***Use of AI tools***

The large language model Claude (Anthropic) was used during this research to help write evaluation harness scripts, draft early versions of manuscript sections, format the LaTeX source for the journal template, refine prose and check bibliographic entries. The authors independently verified all experimental outputs, mathematical proofs and manuscript text, and take full responsibility for the final work.

# 5 Formal analysis

Let $S_w(T)$ be the set of detections emitted by Algorithm 1 for some chunking of $T$, mapped back to absolute offsets, and $P_{w,k}(T)$ the output of the parallel engine with $k$ workers. Each detector $d \in \mathcal{D}$ declares three constants: the longest span it emits, $L_d$, and the context it reads to the left and right of a span, $C_{\text{left},d}$ and $C_{\text{right},d}$. The two assumptions are stated per detector and must hold for every input text, not only for $T$, because the engine applies $\Pi$ to windows.

**Assumption A1** (Bounded span) For every text $X$, every detection of detector $d$ in $\Pi(X)$ has length $j - i \le L_d$.

**Assumption A2** (Bounded context) Let $X$ and $Y$ be texts and $\delta$ an offset with $X[a{:}b] = Y[a{+}\delta : b{+}\delta]$. Let $(i, j)$ be a span of detector $d$ such that either $i - C_{\text{left},d} \ge a$ or $a = a{+}\delta = 0$, and either $j + C_{\text{right},d} \le b$ or both $b = |X|$ and $b + \delta = |Y|$. Then $(i, j, \tau, \kappa) \in \Pi(X)$ if and only if $(i + \delta, j + \delta, \tau, \kappa) \in \Pi(Y)$.

In words, A2 says that whether a span of $d$ is in the final detection set, and with which type and confidence, depends only on the $C_{\text{left},d}$ characters before it, the span itself and the $C_{\text{right},d}$ characters after it, where a window edge that coincides with an edge of both texts counts as real. The pipeline requirement is

$$w^\star \;=\; \max_{d \in \mathcal{D}} \max\big(L_d + C_{\text{right},d},\; C_{\text{left},d}\big). \tag{1}$$

**Theorem 1** (Streaming equivalence) *Under A1 and A2, if $w \ge w^\star$ and $m \ge w$, then $S_w(T) = \Pi(T)$ for every text $T$ and every chunking of $T$.*

**Corollary 1** (Pipeline-wide sufficient condition) *Let $L_{\max} = \max_d L_d$, $C_{\text{left}} = \max_d C_{\text{left},d}$ and $C_{\text{right}} = \max_d C_{\text{right},d}$. Then $w \ge \max(L_{\max} + C_{\text{right}}, C_{\text{left}})$ implies $w \ge w^\star$, so Theorem 1 applies.*

The corollary is the intuition: the overlap must hold the longest span together with the widest right context, and the widest left context. It is sufficient but not necessary, because it pairs the longest span with a right context that may belong to a different detector. Equation (1) is the condition the engine checks (Section 5.1).

***Why the right context is added to the span***

An overlap of $\max_d \max(L_d, C_{\text{left},d}, C_{\text{right},d})$, which bounds each quantity separately, is not enough. Take a detector $d$ that rejects a candidate whose right context is missing, and an overlap $w < L_d + C_{\text{right},d} - 1$. A chunking can end a pass so that a span $s = (i, j)$ of $\Pi(T)$ starts one character before the planned boundary $e_0 = |B| - w$ and has length $L_d$. Its right context then extends $(L_d - 1) + C_{\text{right},d} > w$ characters beyond $e_0$, past the end of the buffer. The window therefore rejects $s$. If no other detection straddles $e_0$, the boundary does not retreat, and the pass emits $T[i : e_0]$, the first character of $s$, unredacted. A shorter span ending before $e_0$ can be emitted in its place. The bound in Eq. (1) is thus tight up to one character. It is also why the safe-region step in the proof below adds $C_{\text{right},d}$ to $L_d$.

*Proof of Theorem 1* Fix a non-final pass. Let $p$ be the absolute position of $B[0]$, $q = p + |B|$ and $e_0 = q - w$ the planned boundary. The window is $W = T[p - |P| : q]$, where $|P| = \min(w, p)$ because $P$ holds the last $w$ emitted characters.

*Safe region.* Let $s = (i, j, \tau, \kappa)$ be a span of detector $d$, in absolute offsets, with $p \le i < e_0$ that belongs to $\Pi(W)$ or to $\Pi(T)$. By A1, $j \le i + L_d < q - w + L_d \le q - C_{\text{right},d}$, using $w \ge L_d + C_{\text{right},d}$. On the left, $i - C_{\text{left},d} \ge p - w$ because $w \ge C_{\text{left},d}$; if $p < w$, the window starts at position 0 of $T$. A2, applied to $X = W$ and $Y = T$, then gives $s \in \Pi(W) \Leftrightarrow s \in \Pi(T)$. So the spans of the window that start in $[p, e_0)$ are exactly the spans of $\Pi(T)$ that start there.

*Retreat.* $\Pi(W)$ is pairwise disjoint, so at most one span $s^*$ straddles $e_0$, and it starts before $e_0$, so it belongs to $\Pi(T)$. The boundary becomes $e = i^*$ if $s^*$ exists and $e = e_0$ otherwise. In the first case, $i^* > e_0 - L_d \ge p + m - L_d \ge p$, because $|B| \ge w + m$ and $m \ge w \ge L_d$. So $e > p$, and the pass emits at least one character; the pinned branch is never taken. No span of $\Pi(T)$ straddles $e$. If $e = e_0$, such a span would start before $e_0$ and so be the straddler of $W$. If $e = i^*$, it would overlap $s^*$, which disjointness rules out.

*Emission.* The pass emits $T[p{:}e]$ with the window spans contained in it. By the safe region these are exactly the spans of $\Pi(T)$ contained in $[p, e)$. The final pass has $q = |T|$, so the right-hand clause of A2 holds for every span, and it emits exactly the spans of $\Pi(T)$ inside $[p, |T|)$. The emitted segments partition $T$, and no span of $\Pi(T)$ crosses a segment boundary. Every span of $\Pi(T)$ is therefore emitted exactly once, with the type and confidence it has in $\Pi(T)$, and nothing else is emitted. □

**Theorem 2** (Parallel identity with fail-safe fallback) *(i) Under the conditions of Theorem 1, and with ranges of at least $4w$ characters, every seam report agrees and $P_{w,k}(T)$ equals the sequential output byte for byte. (ii) Without any assumption on the detectors, the output is either assembled from parts whose seam reports all agree, so that every character of $T$ is emitted exactly once, or it is the sequential output.*

**Theorem 3** (Bounded working buffer) *Let $c$ be the length of the longest input chunk. Under the conditions of Theorem 1, the unemitted buffer of $S_w$ never exceeds $w + m + c$ characters, and each detection window never exceeds $2w + m + c$, independent of $|T|$. Each parallel worker additionally holds back at most $2w$ characters plus one segment, so the parallel engine needs $O\big(k\,(w + m + c)\big)$ characters.*

Because redaction acts on each span separately (Definition 1), Theorem 1 implies byte-identical redacted output. The proofs of Theorems 2 and 3 are in Appendix A,

**Table 1** Declared streaming bounds of the evaluated detectors, in characters, at the evaluated commit. The requirement of a detector is $\max(L_d + C_{\text{right},d},\, C_{\text{left},d})$; the pipeline requirement $w^\star$ is the largest requirement among active detectors.

| Detector $d$ | $L_d$ | $C_{\text{left},d}$ | $C_{\text{right},d}$ | Requirement |
|---|---|---|---|---|
| Deterministic patterns | 8192 | 0 | 0 | 8192 |
| Label-anchored identifiers | 64 | 96 | 8 | 96 |
| Entropy tier | 8192 | 32 | 0 | 8192 |
| Compound entropy tier | 8192 | 0 | 0 | 8192 |
| Contextual scorer | 9 | 48 | 0 | 48 |
| spaCy named-entity tier (optional) | 256 | 256 | 256 | 512 |
| Default pipeline, $w^\star$ | | | | 8192 |
| Default pipeline with spaCy, $w^\star$ | | | | 8192 |

and Online Resource 1 maps each step to the implementation. For `redact_file`, $c = m = 65{,}536$ and $w = 8192$, so the buffer stays below 139,264 characters and each window below 147,456 characters. The proofs were not machine-checked.

## 5.1 What is derived and what is assumed

Each detector declares $(L_d, C_{\text{left},d}, C_{\text{right},d})$. The engine computes $w^\star$ from Eq. (1) over the active detectors and refuses to start with a smaller overlap unless the caller explicitly overrides it. Table 1 lists the declarations. The detectors that emit the longest spans (private-key blocks and entropy candidates, $L_d = 8192$) read no right context. The only detector with right context among the default tiers, the label-anchored tier, emits spans of at most 64 characters. Hence $w^\star = 8192$, and the default overlap $w = 8192$ satisfies Theorem 1 exactly, with or without the spaCy tier. The pipeline-wide Corollary 1 would ask for $8192 + 8 = 8200$ characters; that is why the per-detector form is the one the engine enforces. A regression test fails if any declaration changes, and a differential test probes each detector to check that its real context dependency does not exceed its declaration.

Two parts of A1 and A2 are *not* derived from declarations. They should be read as assumptions corroborated by tests rather than proven properties of the code. (i) The confidence adjustment and threshold must not reach beyond the declared context of the detector whose span they adjust. The contextual scorer's 48-character left window is within $w$ for every detector, but it is not added to the other detectors' declarations. (ii) Overlap resolution must not propagate decisions along chains of overlapping candidates beyond the declared contexts, and the named-entity union rule must not create spans longer than the declared bounds. For the pattern tiers, whose spans are bounded and whose overlaps are short, the equivalence tests in Section 7.1 found no counterexample. A statistical model's output can depend on its whole input window, so for the spaCy tier the declared bounds of 256 characters are declarations only, and the theorems should not be read as covering it. Empirically, its streamed output matched the whole-document output in every chunking we tried (Section 7.1).

### *Failure modes*

Lowering the overlap below $w^\star$ is refused by default. If a caller overrides that, the parallel engine still cannot lose or duplicate characters (Theorem 2(ii)), but the sequential

engine can split a long span. A detector that understates its context, or one that needs document-wide context, is outside the guarantee.

# 6 Experimental methodology

The evaluation addresses seven questions. **RQ1**: do the implementation's streamed, parallel and native outputs match the reference in practice, and how often does chunking without the guarantee leak? **RQ2**: how do memory and throughput scale? **RQ3**: how well do the pattern tiers detect credentials compared with purpose-built scanners? **RQ4**: how well does SafeStream detect personal data compared with Presidio, with and without a named-entity model? **RQ5**: how much protection does it give at the level of a whole document, and how much of that depends on rules written after seeing the benchmark? **RQ6**: how well do the systems detect personal data in real system logs? **RQ7**: how well do they detect personal data in real free text annotated by people other than the authors?

## 6.1 Corpora and splits

Table 2 lists the test splits. Each corpus was split once into development and test halves with a fixed seed. The manifests list document identifiers and the SHA-256 of the source data, and the loader refuses a source that no longer matches. Development, error analysis and tuning used development halves only, with the exception of RedactionBench described below; test halves were scored after the code was frozen, with one disclosed exception (Section 7.4).

- **presidio-research** [15] (synthetic): the dataset distributed with Presidio's evaluation toolkit (revision 0cb3650), with person, location, organisation, telephone, card, date and other labels, generated from templates. The 750 test documents come from 200 templates (median 3.5 documents per template).
- **secrets** (semi-synthetic): the labelled examples in the Nosey Parker rule set [29] (revision 2e6e7f3) as positives, with gold spans given by each rule's own capture groups, plus 200 generated non-secrets (UUIDs, commit hashes, SHA-256 digests, base64 image payloads, minified JavaScript) as gold-free documents. The split is by rule, so the test half contains vendor formats absent from the development half; its 216 positives come from 96 rules. The positives are short examples written by rule authors, not credentials observed in real artefacts.
- **RedactionBench** [33]: 200 documents across eleven domains, a majority seeded from real-world sources according to its authors, with character spans labelled *mandatory* or *contextual*. The test half holds between 5 and 17 documents per domain. All 200 documents were used for error analysis and rule writing before the split existed, so its test half is not held out (Section 7.5).
- **OpenSSH logs** (real): the 2,000-line OpenSSH sample of Loghub [34], real `sshd` lines from a server under attack, one document per line. Gold comes from labels on the 27 log templates that Loghub's parser assigns: the value of a slot that names a remote host is `IP_ADDRESS` when it is a valid IP address (reverse-DNS hostnames are excluded), and the value of a slot that names an account, and any account name

written into a template, is `USERNAME`. Applying the labels gives 1,349 IP addresses and 1,142 user names; 148 lines contain neither. The labels were drafted from the template text with an AI assistant and checked by the first author, and the corpus was frozen as test-only (commit `099cc9c`) before any system was run on it. No SafeStream rule was written with these logs in view.

- **Linux logs** (real, held out): Loghub's 2,000-line Linux sample (sshd authentication through PAM, `su`, `login`, `ftpd`, `klogind` and the kernel), one document per line, labelled under the same rule. That gives 1,258 IP addresses and 618 user names; 407 lines contain neither. Its labels were drafted only after the user-name tier of Section 7.7 had been committed (commit `c699ec6`), checked by the first author, and frozen as test-only (commit `7f87a77`) before any system was run on it, so the corpus is held out from that tier and from every other rule.
- **TAB** (real, human gold, held out): the test split of the Text Anonymization Benchmark [17] (repository commit `558e09e`, MIT licence), 127 English judgments of the European Court of Human Rights annotated by one to ten trained annotators each (555 document–annotator pairs). Annotators marked every mention that identifies the applicant directly (DIRECT) or in combination with other information (QUASI), with types such as person, code, date, organisation and location; these mentions are gold. No rule of the evaluated configurations was written with TAB in view. Its manifest and its scoring protocol (metrics, systems and planned comparisons) were committed before any system was run on it (commit `17b80c4`). TAB's two other splits serve the free-text configuration of Section 7.9: its development split (127 judgments) is the only TAB data used to design it, and its training split (1,014 judgments, 1,112 document–annotator pairs), never inspected or scored before the configuration was frozen, is its held-out confirmation set (protocol committed in `f2be354`).
- **CredData** (real credentials, human gold, held out): Samsung's credential dataset [35] (repository commit `01e3d13`), lines of real files from 337 open-source repositories that its authors labelled as holding a credential, not holding one, or holding a test value or placeholder. Its builder fetches the pinned commits and replaces every real value with a random value of the same shape; we deleted the raw checkouts as soon as it had done so. One document is one labelled line range; a range with a credential carries each credential value as gold, and a range labelled only as non-credential is gold-free. That gives 64,797 documents, 14,626 gold values and 50,608 gold-free ranges; 1,127 credential labels without a value position were dropped by the loader. Only the candidates proposed by its authors' tools, which included gitleaks and TruffleHog, are labelled, so recall is relative to that candidate set. The protocol and the frozen manifest were committed before any system was run on it (commits `f160334` and `8e9e42f`).

## 6.2 Systems and configurations

**SafeStream** was run with the Rust core and default tiers (*pattern*), with the opt-in date tier (*+dates*), and with the spaCy `en_core_web_sm` tier (*+spaCy*). **Presidio** 2.2.355 [4] was run as its pattern recognisers alone (*regex*) and as the full analyser

**Table 2** Test splits. Gold-free documents contain no gold span; any detection on them is a false alarm. Clusters are the resampling units of the cluster and two-stage bootstrap (Section 6.4).

| Corpus | Documents | Gold spans | Gold-free | Bytes | Clusters |
|---|---|---|---|---|---|
| presidio-research | 750 | 1,394 | 63 | 63,678 | 200 templates |
| secrets | 313 | 293 | 97 | 46,595 | 96 rules + 5 templates |
| RedactionBench | 100 | 25,592 | 0 | 1,378,267 | 11 domains |
| OpenSSH logs (real) | 2,000 | 2,491 | 148 | 221,218 | 27 templates |
| Linux logs (real) | 2,000 | 1,876 | 407 | 212,487 | 118 templates |
| TAB judgments (real) | 127 | 6,201[1] | 0 | 652,935 | documents |
| TAB held-out (real) | 1,014 | 59,143[1] | 0 | 8,328,627 | documents |
| CredData (real) | 64,797 | 14,626 | 50,608 | 17,943,839 | 337 repositories |

[1]Union over annotators, overlapping mentions merged; TAB is scored against each annotator's gold separately (Section 6.3).

with spaCy `en_core_web_sm` (*sm*) and `en_core_web_lg` (*lg*), with spaCy 3.8.13 and models 3.8.0. **gitleaks** 8.28.0 [5] and **TruffleHog** 3.90.8 [6] were run with default rules and default false-positive filtering; TruffleHog ran with `--no-verification`, so no example credential was sent to a live service. A sensitivity run repeated both scanners with their false-positive filters disabled (Section 7.3). Each scanner ran once over a directory with one file per document, and findings were mapped back to character offsets; TruffleHog's composite findings (for example `user:token`) were credited for each component that appears in the text. Presidio received documents longer than 20,000 characters in line-aligned pieces of at most that length; no gold span in any test split crosses a piece boundary, so the splitting cost Presidio no recall. All SafeStream thresholds are defaults; no threshold was tuned on test data. The RedactionBench dev-selected configurations (Section 7.5) were chosen on the development half by a rule fixed in advance.

## 6.3 Metrics

Let $G$ and $R$ be the sets of gold and redacted character positions in a document. Character precision and recall are $|G \cap R|/|R|$ and $|G \cap R|/|G|$, summed over documents, and F1 is their harmonic mean. All metrics are type-agnostic: a value counts as protected whatever label it receives. The *fully redacted rate* is the share of gold spans with every character redacted. The *false-alarm document rate* is the share of gold-free documents with at least one detection. On RedactionBench, *mandatory recall* is character recall over mandatory spans, and the *no-leak document rate* is the share of documents in which every mandatory character is redacted (Section 7.5 discusses how it relates to recall). TAB is scored as its authors prescribe [17]: recall against each annotator's gold, pooled over document–annotator pairs; precision counting a redacted character as correct if any annotator marked it; and *entity recall* for direct and quasi identifiers, under which an entity counts as protected only if every one of its mentions is fully redacted. TAB's information-weighted precision needs a language model and is not computed.

### 6.4 Statistical analysis

We report 95% percentile bootstrap intervals with 2,000 resamples and seed 0 [36]. All systems evaluated on a corpus are scored on one shared resample matrix, so every difference between two systems is paired; we report the median paired difference and call it significant only when its paired interval excludes zero. Documents are not independent. Secrets positives cluster by rule, presidio-research documents share generator templates, and RedactionBench documents share domains. We therefore use three resampling schemes on the same per-document counts [37]: *document* (documents drawn independently), *cluster* (whole clusters drawn with replacement, all their documents kept) and *two-stage* (clusters drawn with replacement, then documents drawn with replacement inside each drawn cluster). Table 2 lists the clusters. Tables 4 and 6 give document-level intervals. Table 7 repeats the planned comparisons under all three schemes, and we call a difference *robust* when it is significant under all three. RedactionBench has only eleven clusters, so its cluster intervals are themselves imprecise; we read them as a sensitivity analysis rather than as an exact test. No correction for multiple comparisons was applied; the comparisons in the abstract were planned, the ablation comparisons were not. The analysis was not pre-registered.

### 6.5 Environment

Accuracy runs used a 4-vCPU Intel Xeon virtual machine at 2.80 GHz with 15 GB RAM, Linux and CPython 3.13. Throughput figures come from separate runs, five per configuration, on the same machine with no other benchmark running; the throughput recorded during accuracy runs is not used. Memory was measured as the peak resident set size of a fresh process per point minus its resident set size after imports, so native allocations are included; it is reported in MiB. Re-running every test-split system for the cluster analysis reproduced each point estimate and document-level interval to three decimals. The held-out results come from commit `0765dae`, the results after the entropy fix (Section 7.4) from `3d2c6ac`, and the pre-exposure RedactionBench results from `874e105`, all on the main branch of the public repository.

## 7 Results

### 7.1 RQ1: equivalence, backend parity and leaks caused by chunking

On every test split the Rust core and the pure-Python reference produced identical predictions (presidio-research 178 spans, secrets 298, RedactionBench 3,167, from the frozen commit). The repository's equivalence suite asserts byte-identical streamed, whole-document and parallel output over exhaustive small chunkings, property-based generated inputs, seam-straddling entities and forced fallbacks, and also checks native parity and declared context; it passed 133 of 133 tests at the final commit. These tests corroborate the theorems for the configurations they exercise; they are not proofs, and they do not exercise the named-entity tier's full input dependence.

***How often chunking alone leaks***

The theorems say that chunking cannot add leaks; they do not say how many leaks chunking adds without them. To measure that, we concatenated each test split into one stream, two newlines between documents, and ran the same pattern pipeline in four ways (Table 3; script `benchmarks/chunking_leak.py`):

- *whole*: one pass over the stream, the reference $\Pi(T)$;
- *naive*: fixed chunks of $c$ characters, each detected on its own;
- *overlapping windows*: windows of $c + o$ characters advancing by $c$, each keeping the detections that start in its own $c$ characters, which is common practice; we used $o = 256$ and $o = 8192$, the longest declared span;
- *SafeStream*: Algorithm 1 at its default overlap and the smallest allowed emit size ($m = w$), on five random chunkings per chunk size with chunk lengths drawn uniformly from 1 to $2c$.

We also ran a 4.2 MB log produced by the repository's generator, with 56,153 gold values, as a stand-in for a dense machine log, and the real system logs of Loghub [34]: its 2,000-line samples of 16 systems (SSH, Linux, Apache, Hadoop, HDFS, OpenStack and others; 4.4 MB, repository commit `dd61d09`). Loghub carries no redaction labels, so on it we count only missed reference detections.

Naive chunking leaked gold values that the whole-document pass redacts on every corpus. At $c = 1$ KiB it leaked 237 RedactionBench values, 7 on presidio-research and 5 on secrets, and 872 values in the generated log; at 64 KiB it still leaked 1 RedactionBench value and 12 in the log. A leaked value can be part of a larger gold region, so the number of leaked gold spans can exceed the number of missed detections. Overlapping windows removed most of these leaks, but not all. At 1 KiB they still missed 3 RedactionBench detections and leaked 2 RedactionBench values, 1 presidio-research value and 2 secrets, even with $o = 8192$. In the pattern runs with $o = 8192$, all seven residual misses start 5 to 9 characters into their window and come from detectors that read a cue to their left (a key name, a URL-credential prefix, an account label). The cue lay in the previous window, which these windows discard, so a longer overlap $o$, which only extends a window to the right, cannot recover them. SafeStream keeps exactly this context as its read-only prefix $P$. Both also redacted characters the whole-document pass does not, from candidates cut short at a window edge. The real logs behaved like the test corpora: on Loghub, naive chunking missed 388 of the 18,329 reference detections at 1 KiB and 9 at 64 KiB, and overlapping windows still missed 3 at 1 KiB and 1 at 4 KiB. SafeStream's detection set was identical to the whole-document reference in all 100 pattern-tier runs: five streams, four chunk sizes and five chunkings each. The same held with the spaCy tier on presidio-research, 66 RedactionBench test documents, three Loghub systems and all 127 TAB judgments (real prose, where naive chunking missed 15 detections at 4 KiB), at 4 KiB and 64 KiB with two chunkings each. That last result is an empirical check, not a proof, because the theorems do not cover the spaCy tier (Section 5.1).

**Table 3** Leaks caused by chunking alone. Each cell gives missed reference detections / gold spans leaked by chunking (fully redacted by the whole-document pass, not by the variant); Loghub has no gold, so its cells give missed detections only. SafeStream: runs whose detection set is identical to the whole-document reference, out of the random chunkings tried. Overlapping windows keep the detections that start in their own chunk; $o$ is the window overlap in characters. The generated log is synthetic and dense (56,153 gold values in 4.2 MB). Selected rows; Online Resource 1 (Table S1) gives every corpus at 1, 4 and 64 KiB.

| Corpus (pipeline), chunk $c$ | Naive | $o = 256$ | $o = 8192$ | SafeStream |
|---|---|---|---|---|
| secrets (pattern), 1 KiB | 11 / 5 | 4 / 3 | 2 / 2 | 5 of 5 |
| presidio-research (pattern), 1 KiB | 5 / 7 | 1 / 1 | 1 / 1 | 5 of 5 |
| RedactionBench (pattern), 1 KiB | 74 / 237 | 3 / 2 | 3 / 2 | 5 of 5 |
| RedactionBench (pattern), 64 KiB | 1 / 1 | 0 / 0 | 0 / 0 | 5 of 5 |
| Generated log (pattern), 1 KiB | 957 / 872 | 0 / 0 | 0 / 0 | 5 of 5 |
| RedactionBench, 66 docs (+spaCy), 4 KiB | 20 / 10 | 2 / 0 | 2 / 0 | 2 of 2 |
| Loghub, 16 real systems (pattern), 1 KiB | 388 | 3 | 3 | 5 of 5 |
| Loghub, 16 real systems (pattern), 64 KiB | 9 | 0 | 0 | 5 of 5 |
| Loghub, 3 systems (+spaCy), 4 KiB | 18 | 4 | 4 | 2 of 2 |
| TAB, 127 judgments (+spaCy), 4 KiB | 15 / 2 | 3 / 0 | 3 / 0 | 2 of 2 |

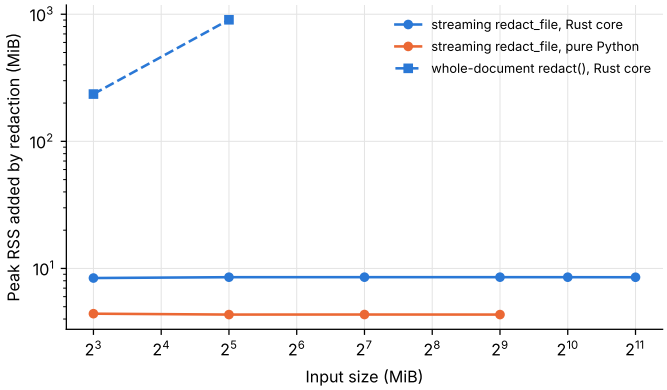



**Fig. 3** Peak resident memory added by redaction (peak RSS minus RSS after imports, MiB) against input size, log scales, each point a fresh process. Streaming `redact_file` stays flat from 8 MiB to 2 GiB (Rust core) and to 512 MiB (pure Python); whole-document `redact()` grows with the input.

## 7.2 RQ2: memory and throughput

Figure 3 plots peak resident memory against input size. Each point is a fresh process redacting a generated log file dense with personal data and credentials. With the Rust core, streaming `redact_file` added between 8.4 and 8.54 MiB at every size from 8 MiB to 2 GiB, a 256-fold range. The pure-Python backend added between 4.3 and 4.4 MiB from 8 to 512 MiB; it was not run at larger sizes because of its run time (726 s at 512 MiB). Whole-document `redact()` on the same generated inputs, measured the same way, added 235.5 MiB at 8 MiB and 904.6 MiB at 32 MiB, about 28 times the input size, because the input string, the detections and the output are held at once; at 32 MiB the two modes therefore differ by a factor of about 106 on identical input. A 128 MiB whole-document run was stopped after 20 minutes without completing and

**Table 4** Test-split results with 95% document-level bootstrap intervals. SafeStream rows are after the entropy fix of Section 7.4, which changed only the secrets results. FR: fully redacted rate; FA: false-alarm document rate.

| Corpus / system | Char P | Char R | Char F1 | FR | FA |
|---|---|---|---|---|---|
| *presidio-research (750 documents)* | | | | | |
| SafeStream pattern | .995 [.992, .998] | .154 [.133, .176] | .267 [.235, .299] | .115 [.098, .134] | .000 |
| SafeStream +dates | .996 [.993, .998] | .169 [.146, .192] | .290 [.255, .322] | .129 [.110, .150] | .000 |
| SafeStream +spaCy (sm) | .898 [.877, .918] | .576 [.547, .606] | .702 [.679, .725] | .538 [.509, .568] | .079 |
| Presidio regex | .948 [.919, .974] | .175 [.151, .202] | .296 [.260, .333] | .123 [.105, .145] | .000 |
| Presidio sm | .871 [.848, .893] | .518 [.491, .546] | .650 [.626, .673] | .506 [.479, .536] | .048 |
| Presidio lg | .878 [.856, .900] | .535 [.507, .563] | .665 [.642, .688] | .562 [.534, .590] | .000 |
| *secrets (313 documents)* | | | | | |
| SafeStream pattern | .857 [.705, .946] | .829 [.740, .887] | .842 [.736, .906] | .437 [.384, .498] | .010 |
| SafeStream +spaCy (sm) | .757 [.604, .858] | .845 [.762, .897] | .799 [.683, .870] | .488 [.435, .548] | .340 |
| gitleaks | .837 [.630, .962] | .473 [.299, .679] | .604 [.430, .765] | .311 [.253, .375] | .000 |
| TruffleHog | .803 [.719, .878] | .189 [.117, .308] | .306 [.204, .446] | .270 [.204, .340] | .000 |
| Presidio regex | .282 [.211, .356] | .105 [.063, .177] | .153 [.098, .225] | .126 [.083, .168] | .392 |
| Presidio sm | .251 [.191, .315] | .119 [.072, .197] | .161 [.107, .232] | .160 [.112, .206] | .649 |
| Presidio lg | .308 [.238, .381] | .131 [.082, .214] | .184 [.124, .262] | .164 [.116, .212] | .433 |

is not reported. These measurements are consistent with Theorem 3 but cover one generated input shape on one machine, and only two whole-document sizes.

On RedactionBench, whose documents average 13.8 kB, the Rust core processed $10.4 \pm 1.3$ MB/s with one thread, the pure-Python reference $1.23 \pm 0.15$ MB/s and Presidio's pattern recognisers 0.069 MB/s; with a statistical model in the loop every system ran below 0.05 MB/s. On a 256 MB generated log, `redact_file` with the Rust core ran at $10.5 \pm 0.5$, $20.6 \pm 0.5$ and $41.1 \pm 1.4$ MB/s with 1, 2 and 4 worker processes, a 3.9-fold speed-up on 4 vCPUs. Online Resource 1 (Table S2 and Fig. S1) gives every corpus and system. These figures come from one virtual machine and should be read as relative rather than absolute.

## 7.3 RQ3 and RQ4: detection on held-out data

Table 4 gives the test-split results (plotted in Online Resource 1, Fig. S2); Table 7 tests the planned differences under clustering.

### *Credentials (RQ3)*

On secrets, SafeStream's pattern tiers fully redacted 0.437 of test secrets, against 0.311 for gitleaks and 0.270 for TruffleHog. With documents resampled independently, the paired differences in fully redacted rate were +0.128 [+0.073, +0.183] and +0.167 [+0.079, +0.250], and in character F1 +0.233 [+0.043, +0.441] and +0.536 [+0.310, +0.685]. Precision did not differ significantly. Resampling by rule widens every interval

**Table 5** Real credentials (CredData): 64,797 labelled line ranges from 337 repositories, values obfuscated by the dataset's builder. FR: fully redacted rate over the 14,626 credential values; FA: share of the 50,608 non-credential ranges with a detection. 95% intervals under two-stage resampling by repository. SafeStream rows: pattern tiers at the evaluated commit, and the current code's default profile.

| System | Char P | Char R | Char F1 | FR / FA |
|---|---|---|---|---|
| SafeStream pattern (evaluated) | .280 [.135, .449] | .817 | .417 [.224, .581] | .548 / .195 [.318, .729 / .105, .298] |
| SafeStream default (current code) | .275 [.135, .450] | .842 | .414 [.229, .590] | .717 / .342 [.551, .832 / .295, .383] |
| gitleaks | .489 [.286, .661] | .630 | .551 [.297, .678] | .398 / .043 [.150, .617 / .008, .097] |
| TruffleHog | .206 | .004 | .008 | .007 / .006 |

(Table 7). The advantage over gitleaks remains significant in both fully redacted rate (two-stage +0.125 [+0.025, +0.209]) and character F1 (+0.226 [+0.011, +0.538]). Against TruffleHog the character-F1 advantage remains (+0.530 [+0.192, +0.742]), but the fully-redacted-rate advantage does not (+0.184 [−0.008, +0.363]); that margin is not robust to dependence between examples of the same rule. SafeStream flagged one of 97 generated non-secrets; neither scanner flagged any. More than half of the test secrets (0.563) were not fully redacted by SafeStream either. The development recall (0.542) exceeded the test recall (0.437) because the test half holds vendor formats for which SafeStream has no pattern. Presidio has no credential recognisers, and adding spaCy raised SafeStream's false-alarm rate to 0.340, mostly from commit words and digests tagged as names on the development split.

### *Scanner sensitivity*

The gold labels are Nosey Parker's rule examples, and both scanners apply false-positive filters designed to suppress example- or placeholder-like strings, so their default configurations may be penalised more than they would be on real data. We therefore re-ran both scanners with these filters disabled, leaving their rules unchanged (Online Resource 1, Section S4 and Table S3). gitleaks gained substantially in character F1 (0.604 to 0.805) but little in fully redacted rate (0.311 to 0.328). Against filters-off gitleaks, SafeStream's fully-redacted-rate advantage remains robust (+0.111 [+0.058, +0.163]; two-stage +0.106 [+0.008, +0.193]), whereas its character-F1 advantage shrinks to +0.039 [+0.008, +0.093] and does not survive rule clustering (+0.039 [−0.007, +0.141]). TruffleHog changed little (fully redacted 0.270 to 0.290, F1 0.306 to 0.314), and the conclusions against it are unchanged. SafeStream's own credential patterns were not developed on the secrets corpus: it was added after the split was fixed, and the only change made using it was the suppression rule of Section 7.4, designed on the development half.

### *Real credentials (CredData)*

The secrets corpus holds rule authors' examples; CredData holds credentials found in real files and labelled by people (Table 5; protocol committed before any run). The planned comparisons show a trade-off rather than a winner. SafeStream's pattern tiers fully redacted more real credentials than gitleaks, 0.548 against 0.398 (+0.150 [+0.143,

+0.157]; two-stage by repository +0.150 [+0.086, +0.279]), so the credential-recall margin of the synthetic evaluation holds on real data and survives clustering. They also raised a false alarm on more of the lines that people judged not to hold a credential, 0.195 against 0.043 (+0.152; two-stage [+0.094, +0.207]), and their character F1 was lower with lines resampled independently (−0.132 [−0.176, −0.095]) but not under clustering ([−0.271, +0.058]). The current code's default profile fully redacted 0.717 of credentials, chiefly because its structured-field tier finds passwords in configuration syntax (fully redacted rate for passwords 0.88, against 0.19 at the evaluated commit and 0.05 for gitleaks), at a false-alarm rate of 0.342. By category, SafeStream led on keys, authorisation headers, URL credentials and nonces, and gitleaks on private keys, tokens and generic secrets (Online Resource 1, Table S7). TruffleHog redacted almost nothing, plausibly because its detectors key on vendor formats that the shape-preserving obfuscation does not keep; we did not verify this. Two readings of the false-alarm gap are possible. The gold-free lines are hard negatives: earlier tools, gitleaks among them, flagged them before people rejected them, so they look like credentials, and flagging one causes over-redaction rather than a leak. But for an alerting scanner the same gap would mean many more alerts to review, which is why gitleaks' design trades recall for precision.

***Personal data (RQ4)***

On presidio-research, the pattern tiers were precise (character precision 0.995) but recalled only 0.154 of sensitive characters, because most gold spans are names, locations and organisations that patterns cannot capture. Against Presidio's pattern recognisers, the paired difference in character recall was −0.021 [−0.041, −0.002] and in precision +0.047 [+0.021, +0.076]; under template clustering the precision advantage remains (+0.046 [+0.011, +0.103]) and the recall deficit is no longer significant. With the same spaCy model, SafeStream +spaCy exceeded Presidio sm in character F1 by +0.051 [+0.027, +0.076] and in fully redacted rate by +0.031 [+0.004, +0.060]. The F1 advantage is robust to template clustering (two-stage +0.052 [+0.013, +0.093]); the fully-redacted-rate advantage is not (+0.033 [−0.014, +0.080]). Against Presidio lg, which uses a larger model, the character-F1 advantage (+0.036 [+0.011, +0.064]) does not survive template clustering, and the fully redacted rates did not differ significantly. This corpus is synthetic and template-generated; the comparison shows relative behaviour on that distribution only.

## 7.4 False alarms on high-entropy non-secrets

Before the suppression rule of Section 4.1, the entropy tier flagged 83 of the 103 generated non-secrets in the development split. The rule, and the secret-key veto that keeps a UUID-shaped access key detected, were designed on the development split. On the test split the false-alarm rate fell from 0.794 [0.714, 0.872] to 0.010 [0.000, 0.035], character precision rose from 0.589 to 0.857, and recall was unchanged. The test half had been scored once with the pre-fix code before the fix was made; both scores are reported. Because the gold-free documents in both halves come from the same generator, the test improvement is an optimistic estimate of the fix's effect on real non-secrets (details in Online Resource 1, Section S5).

**Table 6** RedactionBench test half (100 documents), with 95% document-level bootstrap intervals. *Exposed*: current code, whose rules were written after inspecting all 200 documents. *Pre-exposure*: the last commit before the benchmark entered the repository. *Dev-selected*: configuration chosen on the development half by a fixed rule (maximum development character F1) and scored on the test half once.

| System | Char F1 | Mandatory recall | No-leak docs |
|---|---|---|---|
| Pre-exposure, pattern | .518 [.434, .593] | .589 [.476, .696] | .074 [.022, .129] |
| Pre-exposure, +spaCy | .472 [.395, .547] | .674 [.566, .779] | .137 [.073, .208] |
| Pre-exposure, dev-selected | .469 [.394, .543] | .676 [.568, .781] | .137 [.073, .208] |
| Exposed, pattern | .579 [.490, .662] | .682 [.578, .787] | .084 [.032, .144] |
| Exposed, +spaCy | .512 [.429, .590] | .761 [.657, .860] | .179 [.105, .263] |
| Exposed, dev-selected | .508 [.427, .588] | .762 [.658, .862] | .179 [.105, .263] |
| gitleaks | .214 [.130, .314] | .247 [.140, .372] | .021 [.000, .053] |
| TruffleHog | .030 [.017, .048] | .024 [.013, .040] | .000 |
| Presidio regex | .323 [.250, .393] | .331 [.227, .422] | .011 [.000, .033] |
| Presidio sm | .336 [.265, .402] | .416 [.326, .500] | .084 [.032, .146] |
| Presidio lg | .344 [.277, .409] | .420 [.333, .499] | .105 [.052, .174] |

## 7.5 RQ5: document-level protection and contamination

Table 6 separates the contaminated and uncontaminated results. Rules written after inspecting the benchmark raised test mandatory recall of the pattern tiers by +0.091 [+0.044, +0.163] and character F1 by +0.060 [+0.029, +0.108] relative to the pre-exposure code (paired), and both gains are robust to domain clustering. Because those rules were written while the test documents were visible, this gain is an upper bound on what they contribute to unseen data. One of them, the label-anchored tier, also improved presidio-research, a corpus it was never tuned on (Section 7.10), which is evidence that it generalises. Development-set selection chose the same configuration for both code versions (spaCy on, entropy on, confidence threshold 0.3).

### *Pre-exposure code against the baselines*

The pre-exposure code never saw the benchmark, so it gives the uncontaminated comparison; we re-ran it and every baseline on the test half and scored them on one shared resample matrix (Table 7). With spaCy, its mandatory recall exceeded Presidio lg by +0.262 [+0.091, +0.436] and Presidio sm by +0.265 [+0.093, +0.441], and both advantages remain significant when domains are resampled. Its character-F1 advantage is robust against Presidio sm but not against Presidio lg (two-stage [−0.024, +0.242]). The pattern tiers alone had higher character F1 than either Presidio configuration at the document level, but neither difference survives domain clustering. Against the secret scanners every pre-exposure configuration was ahead under all three schemes. No-leak rates did not differ significantly from Presidio sm or lg.

### *No-leak documents*

The no-leak document rate was low for every system: at most 0.179 [0.105, 0.263] for SafeStream (+spaCy, exposed code), 0.137 for the pre-exposure code with spaCy, 0.084 for the pattern tiers alone, and at most 0.105 for Presidio. In other words, for at least 82% of RedactionBench test documents, every configuration we evaluated left at least one must-redact character in the clear. This metric is deliberately strict. Mandatory recall measures the share of must-redact characters hidden; the no-leak

**Table 7** Planned paired comparisons under three bootstrap schemes: median paired difference and 95% interval (2,000 resamples, seed 0, one shared resample matrix per corpus). † marks an interval that includes zero. Pre-exposure SafeStream on RedactionBench is compared with every baseline because it never saw the benchmark.

| Comparison | Metric | Document | Cluster | Two-stage |
|---|---|---|---|---|
| *secrets: clusters are 96 rules and 5 negative templates* | | | | |
| Pattern − gitleaks | FR | +.128 [+.073, +.183] | +.123 [+.033, +.206] | +.125 [+.025, +.209] |
| Pattern − gitleaks | Char F1 | +.233 [+.043, +.441] | +.215 [+.013, +.527] | +.226 [+.011, +.538] |
| Pattern − TruffleHog | FR | +.167 [+.079, +.250] | +.178† [−.010, +.350] | +.184† [−.008, +.363] |
| Pattern − TruffleHog | Char F1 | +.536 [+.310, +.685] | +.529 [+.192, +.740] | +.530 [+.192, +.742] |
| *presidio-research: clusters are 200 templates* | | | | |
| +spaCy − Presidio sm | Char F1 | +.051 [+.027, +.076] | +.052 [+.013, +.092] | +.052 [+.013, +.093] |
| +spaCy − Presidio sm | FR | +.031 [+.004, +.060] | +.032† [−.015, +.077] | +.033† [−.014, +.080] |
| +spaCy − Presidio lg | Char F1 | +.036 [+.011, +.064] | +.036† [−.000, +.075] | +.037† [−.003, +.080] |
| *RedactionBench, pre-exposure SafeStream: clusters are 11 domains* | | | | |
| +spaCy − Presidio lg | Mand. recall | +.262 [+.091, +.436] | +.252 [+.085, +.348] | +.255 [+.026, +.453] |
| +spaCy − Presidio sm | Mand. recall | +.265 [+.093, +.441] | +.256 [+.104, +.345] | +.259 [+.037, +.454] |
| +spaCy − Presidio lg | Char F1 | +.129 [+.033, +.223] | +.124 [+.006, +.221] | +.123† [−.024, +.242] |
| +spaCy − Presidio sm | Char F1 | +.138 [+.047, +.230] | +.132 [+.031, +.219] | +.132 [+.002, +.242] |
| +spaCy − Presidio lg | No-leak | +.032† [−.042, +.102] | +.036† [−.096, +.157] | +.031† [−.110, +.169] |
| Pattern − Presidio lg | Char F1 | +.176 [+.052, +.289] | +.167† [−.051, +.298] | +.169† [−.068, +.319] |
| Pattern − Presidio lg | Mand. recall | +.176† [−.001, +.353] | +.168† [−.252, +.266] | +.164† [−.239, +.373] |
| Pattern − gitleaks | Mand. recall | +.342 [+.263, +.419] | +.340 [+.217, +.468] | +.335 [+.191, +.479] |
| Pattern − TruffleHog | Mand. recall | +.571 [+.456, +.666] | +.560 [+.304, +.642] | +.563 [+.315, +.677] |
| +spaCy − gitleaks | Mand. recall | +.427 [+.350, +.510] | +.433 [+.320, +.561] | +.439 [+.297, +.589] |
| +spaCy − TruffleHog | Mand. recall | +.656 [+.548, +.749] | +.651 [+.591, +.713] | +.662 [+.526, +.751] |
| *RedactionBench, contamination: exposed minus pre-exposure pattern tiers* | | | | |
| Exposed − pre-exposure | Mand. recall | +.091 [+.044, +.163] | +.094 [+.064, +.159] | +.089 [+.043, +.198] |

rate requires all of them in a document to be hidden, so a single missed character fails the document. RedactionBench documents are dense: the test half averages about 256 gold spans per document (Table 2), of which 84 on average (median 25) are mandatory.

*Where the leaks are.* Per-document predictions of five systems (Online Resource 1, Section S6) show that the leaks are whole missed values rather than boundary errors (4.8% of leaked mandatory characters with spaCy lie in partly redacted spans), that they are concentrated (fixing one or two spans per document would raise the no-leak rate with spaCy only to 0.305), and that they are mostly names without a named-entity model. Even the union of all five systems' redactions would leave 69 of 95 documents leaking (no-leak rate 0.274), so the low document-level rate is a property of

**Table 8** Real OpenSSH logs (2,000 lines; Loghub), all gold mandatory. IP and user-name columns give the gold values fully redacted by any detection, whatever its type. FA: share of the 148 lines without gold that received a detection. Point values; the paired differences in the text carry 95% intervals, including two-stage intervals by template (27 clusters).

| System | Char P | Char R | Char F1 | IP addresses | User names | FA |
|---|---|---|---|---|---|---|
| SafeStream pattern | .730 | .786 | .757 | 1,349 / 1,349 | 0 / 1,142 | .000 |
| SafeStream +spaCy (sm) | .548 | .790 | .647 | 1,349 / 1,349 | 21 / 1,142 | .155 |
| Presidio regex | .661 | .787 | .719 | 1,349 / 1,349 | 4 / 1,142 | .034 |
| Presidio sm | .331 | .790 | .466 | 1,349 / 1,349 | 18 / 1,142 | 1.000 |
| Presidio lg | .421 | .789 | .549 | 1,349 / 1,349 | 14 / 1,142 | .993 |
| gitleaks | .000 | .000 | .000 | 0 / 1,349 | 0 / 1,142 | .000 |
| TruffleHog | .000 | .000 | .000 | 0 / 1,349 | 0 / 1,142 | .000 |

the corpus and of current detectors in general, not only of SafeStream. Nor is it caused by streaming: by Theorem 1 the streamed output equals the whole-document output, so these leaks are detection misses that a whole-document run would also make.

## 7.6 RQ6: detection on real system logs

Table 8 gives the results on real `sshd` lines. Every system with an IP pattern, SafeStream and all three Presidio configurations, redacted all 1,349 IP addresses. No system redacted user names: SafeStream's pattern tiers redacted none of 1,142, and the best configuration, SafeStream with spaCy, redacted 21. User names in phrases such as `Invalid user webmaster from` carry no shape or label that a pattern tier keys on, and the statistical models rarely tag them. This is the clearest gap that real data exposed. Because user names are nearly half the gold, the fully redacted rate is 0.54–0.55 for every IP-capable system, and only 38–40% of lines with gold were fully protected.

The systems differed in what else they redacted. Presidio's statistical configurations tagged the syslog timestamp of almost every line as a date, so they flagged 99–100% of the lines without gold, against none for SafeStream's pattern tiers and 0.155 with spaCy. As a result, SafeStream's character F1 exceeded Presidio sm by +0.291 [+0.286, +0.295] and Presidio lg by +0.208 [+0.202, +0.213], and both differences hold under template clustering (two-stage +0.284 [+0.205, +0.351] and +0.207 [+0.130, +0.275]). Against Presidio's pattern recognisers the character-F1 difference, +0.038 [+0.031, +0.045], does not survive template clustering ([−0.001, +0.122]). One precision caveat applies to every IP-capable system alike. In 385 lines the remote host is a reverse-DNS name that embeds the client's IP address; the frozen labels exclude hostnames, so every system that redacted the embedded address has understated precision. The labels were not changed after scoring. These are one system's logs and two entity types, so they complement rather than replace a broader real-world evaluation (Section 8.4).

## 7.7 A user-name tier for authentication logs

The OpenSSH result exposed a concrete gap, and one that rules can close: in authentication logs, a user name is identified not by its shape but by the fixed text the logging program writes around it. We therefore added an opt-in tier (`use_log_usernames`)

**Table 9** Held-out real Linux logs (2,000 lines; Loghub). Labels were drafted after the user-name tier was committed and frozen before scoring. Columns as in Table 8; FA is over the 407 lines without gold. The upper block is the evaluated commit, as in the rest of the paper; the lower block is the current code with and without the user-name tier.

| System | Char P | Char R | Char F1 | IP addresses | User names | FA |
|---|---|---|---|---|---|---|
| SafeStream pattern | .956 | .866 | .909 | 1,258 / 1,258 | 0 / 618 | .005 |
| SafeStream +spaCy (sm) | .532 | .866 | .659 | 1,258 / 1,258 | 0 / 618 | .452 |
| Presidio regex | .403 | .866 | .550 | 1,258 / 1,258 | 0 / 618 | .251 |
| Presidio sm | .270 | .866 | .412 | 1,258 / 1,258 | 0 / 618 | .526 |
| Presidio lg | .285 | .866 | .429 | 1,258 / 1,258 | 0 / 618 | .686 |
| gitleaks | .000 | .000 | .000 | 0 / 1,258 | 0 / 618 | .000 |
| TruffleHog | .000 | .000 | .000 | 0 / 1,258 | 0 / 618 | .000 |
| Current code, default | .959 | .944 | .952 | 1,258 / 1,258 | 372 / 618 | .005 |
| Current code + user-name tier | .961 | 1.000 | .980 | 1,258 / 1,258 | 618 / 618 | .005 |

with 14 patterns. Each is anchored on the documented message formats of OpenSSH's `sshd` (`Invalid user NAME from`, `Failed password for NAME from`), Linux-PAM's `pam_unix` (`user=NAME`, `session opened for user NAME`), `su`, `sudo` and `login`. Only the name is redacted. The tier declares its bounds ($L = 64$, $C_{\text{left}} = 96$, $C_{\text{right}} = 320$), so $w^\star$ stays 8192, and its tests include the streamed-equals-whole-document check at its own minimal overlap. These results come from the current code, not the evaluated commit, and the current default pipeline already includes a structured-field tier that reads `user=NAME`, so we compare the current code with and without the new tier.

On the held-out Linux logs (Table 9), the tier redacted all 618 user names. Every line with gold was then fully protected (against 0.846 without the tier and 0.612 for every IP-capable evaluated configuration), and the false-alarm rate was unchanged. Against the current default, the gain in fully redacted rate was +0.131 [+0.115, +0.147] with lines resampled independently. Under template clustering its interval reaches zero ([+0.000, +0.771]), because all user names occur in 5 of the 118 templates and a resample can draw none of them. Against every Presidio configuration, the current code with the tier had higher character F1 under template clustering (by at least +0.429 [+0.338, +0.629], against Presidio regex), higher fully redacted rate (+0.347 [+0.024, +0.810]) and more fully protected lines. The evaluated commit's pattern tiers also had higher character F1 than every Presidio configuration under clustering (by at least +0.354 [+0.054, +0.431]), because Presidio flagged identifiers, URLs and timestamps in lines without gold.

On the OpenSSH lines the tier redacted 1,127 of 1,142 user names (386 without it), raising fully protected lines from 0.592 to 0.992 with no false alarms. That result is *exposed*, because those templates had been read while drafting their labels. The 15 misses are the literal name `user`, which the tier treats as a word of the format, and three names logged after a double space, where the gold span includes the space. Both corpora are small and come from two machines, so this is evidence that cues taken from documented formats close the gap in authentication logs, not evidence about user names in general.

**Table 10** TAB test split: 127 real ECHR judgments, scored against each of 555 document–annotator pairs. Gold is every mention marked DIRECT or QUASI. ER: entity recall, the share of entities all of whose mentions are fully redacted. The protocol was committed before any system was run. Character F1 with 95% document-level intervals.

| System | Char P | Char R | Char F1 | ER direct | ER quasi |
|---|---|---|---|---|---|
| SafeStream pattern | .683 | .004 | .008 [.003, .015] | .000 | .002 |
| SafeStream +dates | .978 | .394 | .561 [.532, .594] | .003 | .439 |
| SafeStream +spaCy (sm) | .456 | .325 | .380 [.356, .404] | .071 | .159 |
| Presidio regex | .090 | .034 | .050 [.043, .057] | .000 | .005 |
| Presidio sm | .525 | .680 | .592 [.575, .608] | .067 | .660 |
| Presidio lg | .534 | .697 | .605 [.587, .621] | .077 | .673 |
| gitleaks, TruffleHog | .000 | .000 | .000 | .000 | .000 |

## 7.8 RQ7: personal data in real court judgments

Table 10 gives the results on TAB (by entity type in Online Resource 1, Table S5). The planned comparison with the same spaCy model reverses the synthetic result of Section 7: SafeStream +spaCy had lower character F1 than Presidio sm, −0.213 [−0.243, −0.181], and than Presidio lg, −0.225 [−0.255, −0.193], because it recalled less (−0.355 [−0.391, −0.316] against sm). Most of the gap is dates. Dates are 44.7% of TAB's gold characters; Presidio redacted 0.98 of them, while SafeStream's named-entity tier, which maps only spaCy's person, organisation and place labels, redacted 0.01. SafeStream's opt-in date tier alone redacted 0.88 of date characters at a precision of 0.978 (character F1 0.561), but no evaluated configuration combines it with spaCy, and we did not add one after seeing these results. On the other types the two spaCy-based systems differed in both directions: SafeStream recalled more organisation characters (0.78 against 0.21) and slightly more person-name characters (0.78 against 0.72), Presidio more demographic attributes (0.44 against 0.15). Direct identifiers, 1,011 person-name and 587 code mentions such as application numbers (`36110/97`), were rarely protected by any system: entity recall on direct identifiers was at most 0.077, SafeStream +spaCy and Presidio sm did not differ significantly (+0.004 [−0.008, +0.016]), and in fewer than 1% of document–annotator pairs was every direct identifier redacted. Because an entity counts only if all its mentions are redacted, a name missed at a single mention is unprotected, and application numbers carry no label or checksum that any evaluated tier keys on. The pattern tiers alone recalled almost nothing (0.004), slightly less than Presidio's pattern recognisers (−0.030 [−0.036, −0.024]); the secret scanners flagged nothing. In short, the evaluated configurations were not built for free text, and TAB shows it.

## 7.9 A free-text configuration, confirmed on held-out judgments

The TAB failures have identifiable causes, so we built a configuration for prose and tested it the way the rest of the paper tests rules: designed on development data, frozen, then scored once on data no one had looked at. The protocol, including the systems, metrics and planned comparisons, was committed before any system was

**Table 11** Held-out TAB judgments: the training split (1,014 judgments, 1,112 document–annotator pairs), never inspected or scored before the free-text configuration was frozen. SafeStream free-text: `profile="prose"` with spaCy, frozen before this run. Columns as in Table 10; *Dir. prot.*: share of document–annotator pairs in which every direct identifier is fully redacted. 95% document-level intervals.

| System | Char P | Char R | Char F1 | ER direct | ER quasi | Dir. prot. |
|---|---|---|---|---|---|---|
| SafeStream free-text | .568 | .817 | .670 [.660, .679] | .892 [.878, .906] | .805 [.797, .813] | .787 |
| SafeStream +spaCy (evaluated) | .368 | .360 | .364 [.353, .374] | .120 [.107, .134] | .156 [.150, .163] | .004 |
| Presidio regex | .067 | .038 | .048 [.044, .052] | .009 [.005, .014] | .006 [.005, .007] | .001 |
| Presidio sm | .458 | .625 | .528 [.519, .537] | .112 [.099, .127] | .672 [.661, .683] | .009 |
| Presidio lg | .465 | .641 | .539 [.530, .548] | .129 [.113, .144] | .693 [.682, .703] | .006 |

run on TAB's development or training split (commit `f2be354`). TAB's test split had already been seen and is reported only as a secondary, exposed result. The configuration, `Redactor(profile="prose", use_ner=True)` at commit `9c10975`, adds three general mechanisms, all off in the default profile and all within the declared bounds, so $w^\star$ stays 8192:

- the prose profile's date tier, plus bare years, which the date tier does not match;
- a prose-values tier: sums of money with their currency and number/year references such as case and application numbers (`47676/99`), each matched on word boundaries;
- a prose mode of the spaCy tier that keeps a person's title with the name (spaCy leaves out "Mr" in "Mr Eric Oldham", so the gold mention was only partly redacted) and skips generic institution words such as "the Court" or "Government".

One candidate rule, treating all-capital words as section headings, was tried on the development split and removed because it would also drop acronym organisations, which can identify. No rule names a TAB value.

On the held-out judgments (Table 11) the free-text configuration was ahead of Presidio lg on every planned comparison: character F1 by +0.131 [+0.122, +0.139], character recall by +0.177 [+0.166, +0.187], entity recall on direct identifiers by +0.764 [+0.744, +0.784] and on quasi identifiers by +0.112 [+0.104, +0.122]. Against Presidio sm every margin was larger (character F1 +0.141 [+0.133, +0.149]). It also had higher precision than either (0.568 against 0.465 for lg). The largest change is in direct identifiers: in 78.7% of document–annotator pairs every direct identifier was fully redacted, against at most 0.9% for every other system, because titles now stay with names and application numbers are caught (code recall 0.91 against 0.04 for Presidio lg; by type in Online Resource 1, Table S6). On the development split the configuration reached character F1 0.742; the held-out value, 0.670, is the one to quote. On TAB's exposed test split it reached 0.732 [0.715, 0.748], against 0.605 for Presidio lg. Its streamed output was identical to its whole-document output on held-out judgments in every chunking tried (941,155 characters, the most spaCy accepts in one call; 4 KiB and 64 KiB, two chunkings each), where naive 4 KiB chunking leaked 22 gold mentions and overlapping windows one.

Three limits apply. The configuration was designed on judgments from the same court as the held-out set, so this is evidence for court rulings and similar formal prose, not for free text in general. Its precision is moderate: most remaining false alarms are names of courts and states that TAB's annotators did not mark. And one direct identifier in five document–annotator pairs still leaked somewhere, so the configuration reduces exposure substantially without making redacted judgments safe to publish on its strength alone.

### 7.10 Ablation

An ablation on the test splits (Online Resource 1, Table S4; document-level intervals, not planned) shows that the entropy tier carried most of the credential recall: without it, character F1 on secrets fell from 0.842 to 0.383 (+0.455 [0.242, 0.641]). The label-anchored tier raised character F1 on presidio-research by +0.030 [0.019, 0.044], a corpus it was never tuned on. The contextual scorer's own detections changed no result, and the named-entity union rule raised recall by small, mostly significant amounts (at most +0.013) at no significant cost in F1 on secrets or RedactionBench.

## 8 Discussion

### 8.1 What the results establish, and what they do not

*Established under stated assumptions:* chunked and parallel processing introduce no detection difference relative to the whole-document pipeline, for detectors that respect their declared bounds (Section 5). *Measured:* leaks caused by naive and overlapping chunking on every corpus, including real system logs, and none with SafeStream; flat working memory across a 256-fold range of input sizes; near-linear scaling to four cores; the robust detection margins of Section 7; and, on real logs, complete IP-address redaction, user names redacted by the opt-in tier on held-out data, and no false alarms from the pattern tiers where Presidio's statistical configurations flagged most lines; on real court judgments, lower personal-data recall than Presidio for the evaluated configurations, and, for a free-text configuration designed on development data, higher character F1 and far higher direct-identifier protection than Presidio on 1,014 held-out judgments; on 14,626 real credentials, more fully redacted than gitleaks, with more false alarms on lines people judged harmless. *Not established:* effectiveness on credentials outside CredData's candidate set and file types; document-level protection adequate for public release; robustness to adversarial obfuscation.

### 8.2 When is the approach useful despite a low no-leak rate?

Bounded memory and equivalence do not make redaction complete; they remove one specific failure mode, the leak introduced by chunking itself, and make large inputs processable at all. The value of that depends on the detection coverage for the values that matter in a given deployment. The evidence supports three uses and argues against a fourth.

1. *Defence in depth for structured secrets in machine-generated text.* Where the sensitive values are credentials and structured identifiers with recognisable shapes (logs, configuration dumps, continuous-integration output), the pattern tiers are where coverage is highest in our data, and the guarantee ensures that splitting a multi-gigabyte log does not make coverage worse. Even here more than half of the test secrets were not fully redacted, so the tool reduces exposure rather than eliminating it.
2. *Pre-filtering before human review or controlled sharing.* When output is reviewed, or shared only with parties already bound by agreements, a high-precision filter that reliably processes large inputs reduces the volume of exposed values without being the last line of defence.
3. *Inputs that cannot otherwise be processed.* Whole-document redaction needed about 28 times the input size in memory in our measurements, which rules it out for multi-gigabyte files on ordinary hosts.
4. *Not supported: sole control for public release of free text.* With no-leak rates below 0.18 on RedactionBench and low free-text recall without a named-entity model, the results do not justify releasing redacted free-text documents publicly on the strength of this system alone.

A deployment should set its own acceptance criteria before relying on the system, for example: per-category recall measured on a representative, authorised sample of the target artefacts; zero tolerated leaks for specified credential types on that sample; a review step for document classes with free text; and periodic re-measurement as formats change. None of these was measured here.

## 8.3 Threats to validity

*External validity.* presidio-research is synthetic and template-generated, and its result does not carry over to the real judgments of TAB (Section 7.8); the secrets positives are rule-author examples and the negatives are generated from five templates; RedactionBench is contaminated. The corpora are English-only. The real-log results cover two machines' logs and two entity types, with gold from template labels checked by one author rather than double annotation (a blind second-annotation sheet and agreement script are released for this, `benchmarks/loghub_agreement.py`); they say nothing about credentials in real logs; CredData's labels cover only candidates that other tools proposed, its values are obfuscated, and each document is one line range rather than a whole file; and the user-name tier's OpenSSH result is exposed. TAB is one legal domain, annotated for one person to protect per judgment, and the free-text configuration was designed and confirmed within it; its transfer to other kinds of prose is untested. TAB's test split motivated that configuration and is therefore reported as exposed. *Construct validity.* The secrets gold spans are Nosey Parker's capture groups, so credentials adjacent to a labelled secret (for example a client identifier next to its secret) count as false positives; RedactionBench's contextual labels have low human agreement according to its authors [33]. *Internal validity.* The secrets test half was scored before and after one fix (Section 7.4); RedactionBench was inspected

during development. *Statistical validity.* Eleven RedactionBench domains are few clusters for a bootstrap, so those intervals are approximate; the ablation intervals are document-level only, and no multiplicity correction was applied. *Comparison fairness.* The scanners' filters target example-like strings, which is what the secrets corpus contains; with them disabled, gitleaks closed most of the character-F1 gap (Section 7.3), so that margin should not be claimed. *Environment.* All runs used one virtual machine.

### 8.4 Evaluation plan for real-world evidence

TAB now provides real free text with independent human gold; CredData provides real credentials; the most valuable next evidence is free text from other domains and real credential-bearing text found by a process independent of existing scanners with clear provenance, such as public continuous-integration logs whose licences permit research use or logs from an organisation that authorises the study, annotated by two people with agreement reported. Safeguards: never test discovered credentials against live services, keep raw text encrypted and limited to annotators, report live secrets to their owners, and publish only annotations, hashes and aggregate results. The repository contains a collection plan, annotation guidelines and a scoring pipeline for such a corpus (`benchmarks/realworld/`). Transformer-based detectors would additionally need an empirical streamed-versus-whole-document check, because the theorems do not cover a model whose output depends on its whole input window.

## 9 Conclusion

Chunked redaction need not trade correctness for scale. With detector-declared bounds on span length and context, an overlap that covers every detector's span plus its right context makes the streamed result equal to the whole-document result, a seam cross-check makes the parallel result equal to the sequential one, and the working buffer does not grow with the input. Naive and overlapping chunking leaked values that a whole-document pass redacts, on test corpora and real logs alike, while SafeStream's output matched the whole-document output in every chunking tried; memory stayed flat from 8 MiB to 2 GiB. These properties remove chunking as a source of leaks; they do not make detection complete. Two detection margins survived cluster-robust resampling and the scanner sensitivity run: more credentials fully redacted than gitleaks, which held on 14,626 real credentials (0.55 against 0.40) at a higher false-alarm rate, and, on a synthetic corpus, higher character F1 than Presidio with the same spaCy model. On real court judgments with independent human gold that second margin reversed for the evaluated configuration; a free-text configuration designed on development judgments and frozen before a held-out run then reached character F1 0.67 against 0.54 for Presidio and fully protected the direct identifiers of 79% of held-out document–annotator pairs against at most 1%. In real logs every IP-capable system redacted all IP addresses but none redacted user names, a gap that an opt-in tier built from documented log formats closed on held-out Linux logs. Other margins did not survive, and fewer than one RedactionBench document in five was fully protected by any configuration. The system is therefore suited to reducing exposure in large machine-generated artefacts as one layer among several, not to certifying

documents for public release. The next step is evaluation on authorised real-world text (Section 8.4).

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## Declarations

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- **Competing interests.** The authors declare no competing financial or non-financial interests.
- **Ethics approval and consent to participate.** Not applicable: no human participants were involved and the authors collected no data. presidio-research is synthetic, the secrets corpus consists of example strings from an open-source rule set together with generated negatives, and RedactionBench is a public dataset released by its authors under CC BY 4.0, which they describe as seeded from real-world sources. All processing was local; no excerpt of any document is published here, and no credential was sent to a live service (TruffleHog ran without verification).
- **Consent for publication.** Not applicable.
- **Data availability.** presidio-research, Nosey Parker and Loghub are public (Loghub is free for research use with citation of [34]); RedactionBench is available from its authors under CC BY 4.0. Generated non-secrets are produced by the released code with a fixed seed. Split manifests, raw results and per-document predictions (character offsets only, no corpus text) are in the public repository under `research/results/`.
- **Code availability.** Source code and the evaluation harness, including the scripts `benchmarks/cluster_bootstrap.py` and `benchmarks/scanner_sensitivity.py`, are available at https://github.com/MounishSenisetty/SafeStream-Redactor.
- **Author contributions.** Mounish Senisetty: conceptualisation, methodology, software architecture, formal analysis, validation, investigation, data curation, visualisation, writing (original draft) and writing (review and editing). Ashrith Reddy N: software implementation, validation, investigation and writing (review and editing). Sandhiya Kennedy: software implementation, validation, investigation and writing (review and editing). Parvathy R: supervision and writing (review and editing). All authors read and approved the final manuscript.
- **Use of generative AI.** During the preparation of this work the authors used the large language model Claude (Anthropic) to help implement evaluation scripts, draft early manuscript text, convert LaTeX markup to the journal template, edit language and check bibliographic references. After using this tool, the authors reviewed

and revised all code, experimental results and text, and take full responsibility for the content.

# Appendix A Proofs of Theorems 2 and 3

Online Resource 1 (Table S8) maps each step below to the code that discharges it and the tests that corroborate it. Throughout, positions are absolute offsets in $T$, the conditions of Theorem 1 hold unless stated otherwise, and $L_{\max} = \max_d L_d \le w$.

## A.1 A streaming engine started mid-text

The parallel engine runs Algorithm 1 on a substring of $T$ with a read-only left context. Lemma 1 covers that case.

**Lemma 1** *Let $X = T[x_0{:}x_1]$ and let the engine start with $P = T[x_0{:}x_0 + h]$ as already-emitted context and stream $T[x_0 + h : x_1]$, where $h = w$, or $h < w$ and $x_0 = 0$. Then it emits exactly the spans of $\Pi(X)$ that start at or after $x_0 + h$.*

*Proof* The proof of Theorem 1 uses only two properties of $P$: it holds the $\min(w, p - x_0)$ characters preceding $B$, and the window ends at the end of the text being processed. Both hold here with $X$ in place of $T$. Spans of the first window that start inside $P$ are dropped, as in every later pass. □

**Lemma 2** (Transfer to the full text) *With $X = T[x_0{:}x_1]$, let $s$ be a span of detector $d$ that belongs to $\Pi(X)$ or to $\Pi(T)$, with $i - C_{\text{left},d} \ge x_0$ or $x_0 = 0$, and $j + C_{\text{right},d} \le x_1$ or $x_1 = |T|$. Then $s \in \Pi(X)$ if and only if $s \in \Pi(T)$.*

*Proof* Apply A2 with $a = x_0$ and $b = x_1$ in $T$; in $X$ these positions are its two edges. This gives $\delta = -x_0$ in the notation of A2, and an edge of $X$ that coincides with an edge of $T$ satisfies the edge clause. □

## A.2 Proof of Theorem 2

Let the split points be $0 = a_0 < a_1 < \cdots < a_k = |T|$, with $a_{r+1} - a_r \ge 4w$; `plan_ranges` enforces this, and coarsens the split or falls back to one range otherwise. Worker $r$ processes $X_r = T[x_0^r : x_1^r]$ with $x_0^r = \max(0, a_r - 2w)$ and $x_1^r = \min(|T|, a_{r+1} + w)$. Its first $w$ characters, $T[a_r - 2w : a_r - w]$, are read-only context, and streaming starts at $a_r - w$; worker 0 has no context and starts at 0. By Lemma 1, worker $r$ obtains $D_r$, the spans of $\Pi(X_r)$ that start at or after $a_r - w$.

A *straddler* of a seam $a$ is a span with $i < a < j$. Worker $r$ reports the straddler of $a_r$ (its start seam) and the straddler of $a_{r+1}$ (its end seam) found in $D_r$, or reports that none exists. It emits $T[\ell_r : u_r]$, where $\ell_r$ is the end of its start-seam straddler or $a_r$, and $u_r$ is the end of its end-seam straddler or $a_{r+1}$. Together with that text it emits the spans of $D_r$ contained in $[\ell_r, u_r)$.

*Step 1: in the window that matters, worker $r$ sees $\Pi(T)$.* Let $s$ be a span of detector $d$ with $a_r - w \le i < a_{r+1}$ that belongs to $D_r$ or to $\Pi(T)$. On the left, $i - C_{\text{left},d} \ge a_r - 2w \ge x_0^r$, because $C_{\text{left},d} \le w$. On the right, $j \le i + L_d < a_{r+1} + L_d \le a_{r+1} + w - C_{\text{right},d}$, so $j + C_{\text{right},d} \le x_1^r$, or $x_1^r = |T|$. By Lemma 2, $s \in D_r \Leftrightarrow s \in \Pi(T)$.

*Step 2: both neighbours report the same straddler.* A straddler of $a_r$ starts after $a_r - L_d \ge a_r - w$, and a straddler of $a_{r+1}$ starts before $a_{r+1}$, so both lie in the range of Step 1. Spans that start before $a_r - w$ are dropped as context. Such a span ends at most $L_{\max} \le w$ characters after its start, that is, at or before $a_r$, so it cannot cross $a_r$. Each straddler found by worker $r$ is therefore the unique straddler of $\Pi(T)$ at that seam, if one exists (uniqueness follows from disjointness). Worker $r+1$ applies Step 1 to its own start seam $a_{r+1}$ and obtains the same span. The reports agree, so the valve passes, and $u_r = \ell_{r+1}$.

*Step 3: each part is the sequential redaction of its slice.* Every span emitted by worker $r$ lies in $[\ell_r, u_r)$. It starts at or after $\ell_r \ge a_r$, and before $a_{r+1}$: if there is an end-seam straddler, any span starting at or after $a_{r+1}$ and ending by $u_r$ would overlap it; otherwise $u_r = a_{r+1}$. By Step 1 the emitted spans are exactly the spans of $\Pi(T)$ inside $[\ell_r, u_r)$. No cut point $\ell_r$ lies strictly inside a span of $\Pi(T)$, because it is either a seam with no straddler or the end of a span of $\Pi(T)$, and $\Pi(T)$ is disjoint. The slices $[\ell_r, u_r)$ therefore partition $T$ without cutting any span. Redaction acts on each span separately (Definition 1), so the concatenated parts equal the redaction of $T$ under $\Pi(T)$. By Theorem 1 that is the sequential output.

*Step 4: timing of the cuts.* A worker writes text only once the cuts that can affect it are fixed. It decides $\ell_r$ once its stream has passed $a_r + w$. By then every straddler of $a_r$, which ends before $a_r + L_{\max}$, has been emitted by the streaming engine. It flushes only text below the current position minus $2w$. The stream never extends beyond $a_{r+1} + w$, so this text lies below $a_{r+1} - w$, and the end cut $u_r \ge a_{r+1}$ cannot affect it. This proves (i).

*Proof of (ii).* This part uses no assumption on the detectors. For each seam, the coordinator compares the left worker's end report with the right worker's start report and redacts the whole file sequentially on any mismatch. If all reports match, each $u_r$ equals $\ell_{r+1}$, because both are computed from the same reported straddler end. The written parts then cover $[0, |T|)$ as consecutive, non-overlapping slices, and every character of $T$ is emitted exactly once, redacted or not. A non-splittable text encoding, a single worker or a small file also take the sequential path. When the conditions of Theorem 1 fail, agreeing parts can still contain different detections from the sequential output; this is why an overlap below $w^\star$ is refused by default. □

## A.3 Proof of Theorem 3

*After an emission.* The retained buffer is $B[e{:}]$. If no span straddles the planned boundary, $|B| - e = w$. Otherwise $e = i^* > |B| - w - L_d$, so $|B| - e < w + L_{\max}$. In both cases the retained buffer holds fewer than $w + L_{\max} \le w + m$ characters.

*Before a detection pass.* Chunks are appended until $|B| \ge w + m$. Before the last of them, $|B| < w + m$, and that chunk adds at most $c$ characters, so a pass starts with $|B| < w + m + c$. The proof of Theorem 1 shows that the pinned branch, which would let $B$ grow further, is never taken when $m \ge w \ge L_{\max}$. The final buffer obeys the

same bound. The window $P \cdot B$ adds $|P| \leq w$, which gives $2w+m+c$. The detections of a window are pairwise disjoint and lie inside it, so their number is at most its length.

*Parallel workers.* Each worker runs one streaming engine and holds back emitted segments until they lie $2w$ behind its stream position, together with the detections of that suffix. This adds at most $2w$ characters plus one segment of at most $w + m + c$ characters. The bound does not depend on $|T|$, and $k$ workers need at most $k$ times as much. □

# References


[1] Alneyadi, S., Sithirasenan, E., Muthukkumarasamy, V.: A survey on data leakage prevention systems. Journal of Network and Computer Applications **62**, 137–152 (2016) https://doi.org/10.1016/j.jnca.2016.01.008

[2] Meli, M., McNiece, M.R., Reaves, B.: How bad can it git? Characterizing secret leakage in public GitHub repositories. In: Network and Distributed System Security Symposium (NDSS) (2019). https://doi.org/10.14722/ndss.2019.23418

[3] Krause, A., Klemmer, J.H., Huaman, N., Wermke, D., Acar, Y., Fahl, S.: Pushed by accident: A mixed-methods study on strategies of handling secret information in source code repositories. In: 32nd USENIX Security Symposium, pp. 2527–2544 (2023). https://www.usenix.org/conference/usenixsecurity23/presentation/krause

[4] Microsoft: Presidio: Data Protection and De-identification SDK. https://github.com/microsoft/presidio. Version 2.2.355 evaluated. Accessed 11 October 2026 (2026)

[5] Gitleaks contributors: Gitleaks. https://github.com/gitleaks/gitleaks. Version 8.28.0 evaluated. Accessed 11 October 2026 (2026)

[6] Truffle Security: TruffleHog. https://github.com/trufflesecurity/trufflehog. Version 3.90.8 evaluated. Accessed 11 October 2026 (2026)

[7] Basak, S.K., Cox, J., Reaves, B., Williams, L.: A comparative study of software secrets reporting by secret detection tools. In: 2023 ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM) (2023). https://doi.org/10.1109/ESEM56168.2023.10304853

[8] Ptacek, T.H., Newsham, T.N.: Insertion, evasion, and denial of service: Eluding network intrusion detection. Technical report, Secure Networks, Inc. (January 1998)

[9] Wang, X., Hong, Y., Chang, H., Park, K., Langdale, G., Hu, J., Zhu, H.: Hyperscan: A fast multi-pattern regex matcher for modern CPUs. In: 16th USENIX Symposium on Networked Systems Design and Implementation (NSDI), pp. 631–648 (2019). https://www.usenix.org/conference/nsdi19/presentation/wang-xiang

[10] Intel Corporation: Introduction to Hyperscan. https://www.intel.com/content/www/us/en/developer/articles/technical/introduction-to-hyperscan.html. Accessed 11 October 2026 (2019)

[11] Neamatullah, I., Douglass, M.M., Lehman, L.-w.H., Reisner, A., Villarroel, M., Long, W.J., Szolovits, P., Moody, G.B., Mark, R.G., Clifford, G.D.: Automated de-identification of free-text medical records. BMC Medical Informatics and Decision Making **8**, 32 (2008) https://doi.org/10.1186/1472-6947-8-32

[12] Norgeot, B., Muenzen, K., Peterson, T.A., Fan, X., Glicksberg, B.S., Schenk, G., Rutenberg, E., Oskotsky, B., Sirota, M., Yazdany, J., Schmajuk, G., Ludwig, D., Goldstein, T., Butte, A.J.: Protected health information filter (Philter): accurately and securely de-identifying free-text clinical notes. npj Digital Medicine **3**, 57 (2020) https://doi.org/10.1038/s41746-020-0258-y

[13] Stubbs, A., Uzuner, Ö.: Annotating longitudinal clinical narratives for de-identification: The 2014 i2b2/UTHealth corpus. Journal of Biomedical Informatics **58**(Suppl), 20–29 (2015) https://doi.org/10.1016/j.jbi.2015.07.020

[14] Dernoncourt, F., Lee, J.Y., Uzuner, Ö., Szolovits, P.: De-identification of patient notes with recurrent neural networks. Journal of the American Medical Informatics Association **24**(3), 596–606 (2017) https://doi.org/10.1093/jamia/ocw156

[15] Microsoft: presidio-research: Toolbox for developing and evaluating PII detectors. https://github.com/microsoft/presidio-research. Synthetic dataset `synth_dataset_v2.json`, revision 0cb3650. Accessed 11 October 2026 (2026)

[16] Lison, P., Pilán, I., Sánchez, D., Batet, M., Øvrelid, L.: Anonymisation models for text data: State of the art, challenges and future directions. In: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers), pp. 4188–4203 (2021). https://doi.org/10.18653/v1/2021.acl-long.323

[17] Pilán, I., Lison, P., Øvrelid, L., Papadopoulou, A., Sánchez, D., Batet, M.: The text anonymization benchmark (TAB): A dedicated corpus and evaluation framework for text anonymization. Computational Linguistics **48**(4), 1053–1101 (2022) https://doi.org/10.1162/coli_a_00458

[18] Tjong Kim Sang, E.F., De Meulder, F.: Introduction to the CoNLL-2003 shared task: Language-independent named entity recognition. In: Proceedings of the Seventh Conference on Natural Language Learning at HLT-NAACL 2003, pp. 142–147 (2003). https://aclanthology.org/W03-0419

[19] Devlin, J., Chang, M.-W., Lee, K., Toutanova, K.: BERT: Pre-training of deep

bidirectional transformers for language understanding. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), pp. 4171–4186 (2019). https://doi.org/10.18653/v1/N19-1423

[20] Sanh, V., Debut, L., Chaumond, J., Wolf, T.: DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. arXiv:1910.01108 (2019). https://doi.org/10.48550/arXiv.1910.01108

[21] Zaratiana, U., Tomeh, N., Holat, P., Charnois, T.: GLiNER: Generalist model for named entity recognition using bidirectional transformer. In: Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), pp. 5364–5376 (2024). https://doi.org/10.18653/v1/2024.naacl-long.300

[22] Krawczyk, H., Bellare, M., Canetti, R.: HMAC: Keyed-Hashing for Message Authentication. RFC 2104 (1997). https://doi.org/10.17487/RFC2104

[23] Sweeney, L.: k-anonymity: A model for protecting privacy. International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems **10**(5), 557–570 (2002) https://doi.org/10.1142/S0218488502001648

[24] Sinha, V.S., Saha, D., Dhoolia, P., Padhye, R., Mani, S.: Detecting and mitigating secret-key leaks in source code repositories. In: 2015 IEEE/ACM 12th Working Conference on Mining Software Repositories (MSR), pp. 396–400 (2015). https://doi.org/10.1109/MSR.2015.48

[25] Saha, A., Denning, T., Srikumar, V., Kasera, S.K.: Secrets in source code: Reducing false positives using machine learning. In: 2020 International Conference on COMmunication Systems & NETworkS (COMSNETS) (2020). https://doi.org/10.1109/COMSNETS48256.2020.9027350

[26] Feng, R., Yan, Z., Peng, S., Zhang, Y.: Automated detection of password leakage from public GitHub repositories. In: Proceedings of the 44th International Conference on Software Engineering (ICSE) (2022). https://doi.org/10.1145/3510003.3510150

[27] Basak, S.K., Neil, L., Reaves, B., Williams, L.: SecretBench: A dataset of software secrets. In: 2023 IEEE/ACM 20th International Conference on Mining Software Repositories (MSR) (2023). https://doi.org/10.1109/MSR59073.2023.00053

[28] Shannon, C.E.: A mathematical theory of communication. Bell System Technical Journal **27**(3), 379–423 (1948) https://doi.org/10.1002/j.1538-7305.1948.tb01338.x

[29] Praetorian: Nosey Parker. https://github.com/praetorian-inc/noseyparker. Rule set at revision 2e6e7f3. Accessed 11 October 2026 (2026)

[30] Muthukrishnan, S.: Data streams: Algorithms and applications. Foundations and Trends in Theoretical Computer Science **1**(2), 117–236 (2005) https://doi.org/10.1561/0400000002

[31] Thompson, K.: Regular expression search algorithm. Communications of the ACM **11**(6), 419–422 (1968) https://doi.org/10.1145/363347.363387

[32] Luhn, H.P.: Computer for Verifying Numbers. US Patent 2,950,048 (1960)

[33] Brynjólfsson, S., Jayakrishnan, S., Sali, E., Purwar, D., Aggarwal, M.: RedactionBench. arXiv:2606.18782 (2026). https://doi.org/10.48550/arXiv.2606.18782

[34] Zhu, J., He, S., He, P., Liu, J., Lyu, M.R.: Loghub: A large collection of system log datasets for AI-driven log analytics. In: 2023 IEEE 34th International Symposium on Software Reliability Engineering (ISSRE), pp. 355–366 (2023). https://doi.org/10.1109/ISSRE59848.2023.00071

[35] Yun, J., Choi, S., Lee, Y., Sokol, O., Shim, W., Melkonyan, A., Kuzmenko, D.: Project CredData: A Dataset of Credentials for Research. https://github.com/Samsung/CredData. Samsung; repository commit 01e3d13 (2021)

[36] Efron, B., Tibshirani, R.J.: An Introduction to the Bootstrap. Chapman & Hall/CRC, New York (1993). https://doi.org/10.1201/9780429246593

[37] Field, C.A., Welsh, A.H.: Bootstrapping clustered data. Journal of the Royal Statistical Society: Series B (Statistical Methodology) **69**(3), 369–390 (2007) https://doi.org/10.1111/j.1467-9868.2007.00593.x